
Fo Guang — Buddha’s Light

Turning the Surveillance State into a Public Good
as a Resource for Rapid Response

A position and systems paper building on ABot-Recon

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ABSTRACT

Cities have already paid for the sensors. Hundreds of millions of fixed CCTV cameras, traffic cameras, transit cameras, doorbell cameras and body-worn cameras record the built environment continuously — yet in the minutes after an earthquake, flood, structural collapse or wildfire, responders still operate on paper maps, phone calls and whatever a drone can capture once one arrives. We argue that the marginal capability gap between “surveillance state” and “civic rapid-response infrastructure” is no longer sensing; it is *geometry at the edge*. Fo Guang (Fó Guāng, “Buddha’s light”) is a proposal and reference design that repurposes existing camera streams into live, metrically consistent 3D reconstructions for emergency response, built on ABot-Recon, a streaming 3D reconstruction model with a fixed 12-frame local context window that produces camera trajectories, dense point maps and confidence estimates from ordinary monocular video.

The pivotal enabler — and the core technical contribution we highlight — is ABot-Recon’s **CPU inference path**. Surveillance infrastructure is overwhelmingly GPU-free: DVRs, NVRs, municipal edge boxes and donated commodity hardware. We describe the engineering that makes a 1.0-billion-parameter geometry transformer run end-to-end on such hardware: a numerically bit-identical pure-PyTorch 2D rotary position encoding replacing a CUDA-only kernel, device-aware attention-backend resolution with a paged-KV cache that degrades gracefully from FlashInfer to native SDPA, device-agnostic autocast, and a *selectively scoped* dynamic INT8 quantisation path whose per-submodule accuracy ablation shows why naive whole-model quantisation destroys geometric output (~100% relative L2 on point maps) while a measured scope costs under 1%. On 8 commodity x86 cores the full pipeline runs at ~0.31 FPS — roughly 80× slower than the 24.45 FPS H100 baseline, yet fast enough for the after-action and slow-scan regimes that dominate civil protection, and deployable on hardware that already sits in every camera closet. We close with a governance design — dual-key activation, geometry-only egress and public audit — under which surveillance optics become a public good without becoming a deeper surveillance state.

Keywords: streaming 3D reconstruction · CPU inference · post-training quantisation · edge computing · disaster response · civic infrastructure · privacy by architecture

1 Introduction

1.1 The stranded asset

The world’s installed camera base is the largest distributed sensor network ever built, and it is almost entirely *stranded* as a civic asset. Its output is consumed by three narrow pipelines: live human monitoring,

forensic playback after crimes, and — increasingly — person-centric analytics such as face recognition, re-identification and behaviour flagging, the pipeline that earns the label “surveillance state.” None of these produce the artefact that rapid response actually needs: an up-to-date, metrically consistent 3D model of the environment. Which streets are passable. Which façades have shifted. Where debris fields begin. How floodwater is propagating.

That artefact used to require survey crews, LiDAR trucks or photogrammetry flights measured in days. The recent generation of feed-forward geometry transformers — DUS3R-lineage models such as Pi3 [2, 3], and streaming systems such as ABot-Recon [1] built upon them — collapse it to a forward pass: monocular video in, camera trajectory and dense point maps out, no calibration targets, no structure-from-motion pipeline, no depth sensor.

1.2 Why the bottleneck is CPU

There is a hidden assumption in nearly all of this literature: that an NVIDIA GPU is available. ABot-Recon’s published efficiency figure — 24.45 FPS at 504×280 on an H100 — is impressive precisely because the H100 is a \$30,000 data-centre part. Municipal camera infrastructure contains no H100s. It contains:

- NVRs and DVRs built around low-power x86 or ARM SoCs;
- traffic-management cabinets with fanless industrial PCs;
- emergency-operations-centre workstations bought on five-year refresh cycles;
- in the developing world and in disaster zones, *whatever laptop survived*.

A civic rapid-response capability that requires shipping GPUs into a disaster area is not a capability; it is a procurement programme. The deciding question for Fo Guang is therefore not “how fast is the model on an H100?” but “does it run *at all*, correctly, on the machines that already exist next to the cameras?” Until recently the answer for ABot-Recon — as for most of its peers — was no: the model could not even be *imported* without a compiled CUDA extension.

This is worth stating plainly, because it is the kind of defect that does not appear in any accuracy table. A model whose published numbers are excellent and whose import statement fails on 99% of the world’s deployed computers has an availability of zero on that hardware. No amount of additional accuracy changes that number. The work of [Section 3](#) moves it from zero, and everything in [Sections 5](#) and [6](#) follows from that move.

1.3 Contributions

1. **A systems account of CPU enablement** ([Section 3](#)): the concrete engineering that takes a CUDA-native streaming geometry transformer to a correct, tested, end-to-end CPU path — RoPE kernel replacement with bit-identical semantics, attention-backend resolution, paged-KV fallback and device-agnostic mixed-precision handling — together with the design rules we extracted for other models in this class.
2. **A quantisation ablation for geometry models** ([Section 4](#)): per-submodule dynamic INT8 measurements against an FP32 baseline showing a sharp cliff. Quantising the point decoder or the shared decoder blocks moves local point maps by $\sim 100\%$ relative L2 — that is, destroys them — while a selective scope costs 2.1 mm mean ATE, 0.68° mean rotation and under 1% on point and confidence maps. We argue that *scope selection by measured geometric regression*, not layer count, is the correct quantisation methodology for reconstruction models.
3. **The Fo Guang architecture and governance design** ([Sections 5](#) and [6](#)): a tiered deployment model mapping reconstruction regimes — live, slow-scan, after-action — onto the hardware actually present

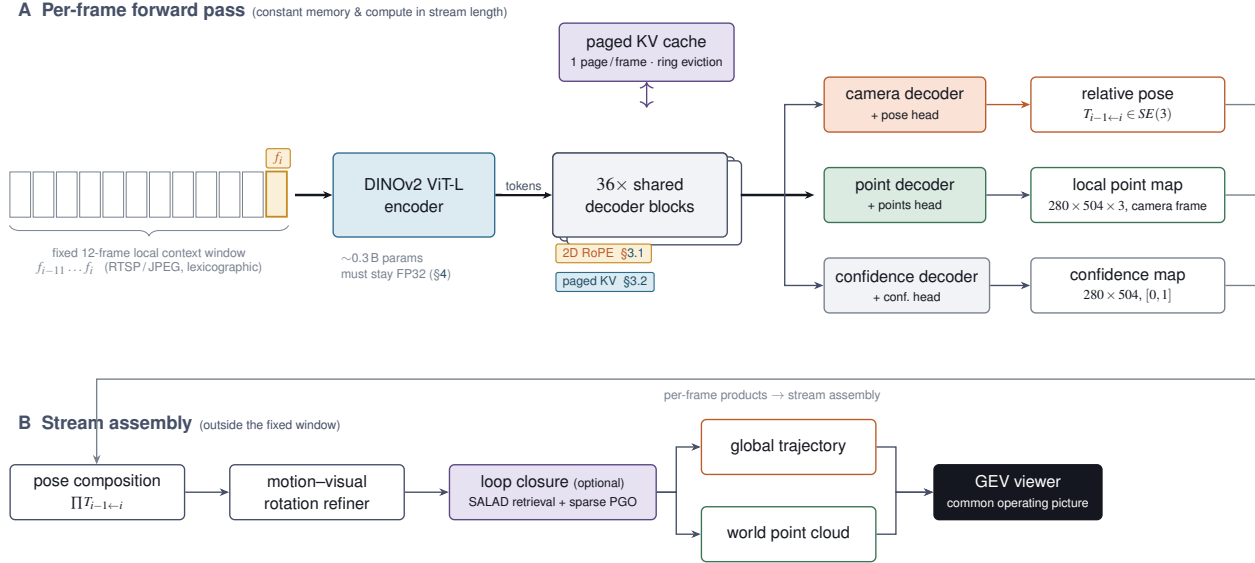


Figure 1: ABot-Recon streaming architecture. A DINOv2 ViT-L encoder embeds each arriving frame; 36 shared transformer decoder blocks with 2D rotary position encoding attend over a *fixed* 12-frame local context window held in a paged KV cache, and three task-specific decoder/head pairs emit a dense local point map, a relative pose, and a per-pixel confidence map. Because the window is fixed, per-frame memory and compute are constant in stream length — the property that makes unbounded, always-on civic operation thinkable at all. Row B composes those per-frame products into a global trajectory and world point cloud. The two shaded call-outs mark the sites that were CUDA-only before the work of Section 3.

in camera networks, and a dual-key, geometry-only-egress governance framework under which the capability is a public good rather than a surveillance deepening.

2 Background: ABot-Recon

ABot-Recon [1] is a streaming 3D reconstruction model derived from the Pi3 architecture [2]. A DINOv2 ViT-L encoder [4] embeds each incoming frame; 36 shared transformer decoder blocks with 2D rotary position encoding [7] attend over a **fixed 12-frame local context window** held in a KV cache; task-specific decoders and heads emit, per frame:

- a dense **local point map** (per-pixel 3D in camera coordinates, 280×504),
- a **relative pose** $T_{i-1 \leftarrow i}$ to the previous frame,
- a per-pixel **confidence map** in $[0, 1]$.

Global trajectories and world point clouds are obtained by composing relative poses; a motion-visual rotation refiner suppresses drift, and an optional loop-closure stage (DINOv2-SALAD retrieval [5] plus sparse pose-graph optimisation) refines long sequences with revisits. Because the context window is fixed, memory and compute are constant in stream length — the property that makes unbounded, always-on civic operation thinkable at all. Figure 1 summarises the architecture.

Total parameters: ~ 1.0 B. Released checkpoint: ~ 4.0 GB of FP32 SafeTensors. Reference throughput: 24.45 FPS at 6.71 GiB peak, 504×280 , on an H100, using FlashInfer paged-KV attention [6] and a compiled cuRoPE CUDA kernel.

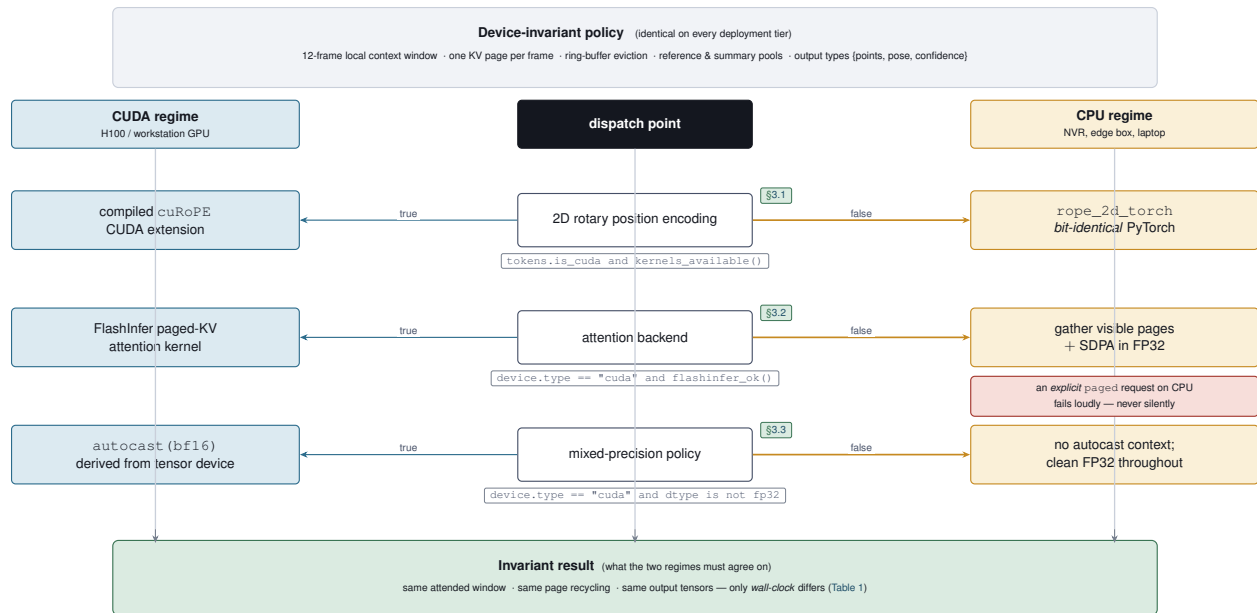


Figure 2: The CPU path is a dispatch layer, not a port. Three sites decided the model’s device by configuration rather than by data, and each made the package CUDA-mandatory. Correcting them leaves one codebase serving both regimes: the *policy* that governs what the model may attend to and what it must emit is device-invariant (top and bottom bands), while only the *kernel* that executes it dispatches on device. The single deliberate asymmetry is the loud failure at lower right — silent fallback would misreport a benchmarked configuration.

3 The CPU Optimisation Innovation

The CPU path is not a port; it is a set of *dispatch corrections* that let one codebase serve both regimes (Fig. 2). Three families of defect made the package CUDA-mandatory, and each has a general form worth naming. We describe each blocker, its fix, and the rule we extracted.

3.1 Rotary position encoding: a bit-identical PyTorch fallback

Blocker. The 2D RoPE module imported its compiled CUDA extension unconditionally; on any machine without the built kernel, the package failed at import time. The GPU-optional feature was, in practice, GPU-mandatory.

Fix. The extension import becomes optional (`_kernels = None` on failure), and the forward pass dispatches per call:

```
if not (tokens.is_cuda and kernels_available()):
    return rope_2d_torch(tokens, positions, self.base, self.F0)
```

The subtlety is *numerical identity*. A generic PyTorch RoPE would be merely “close” to the kernel; geometric models compound pose over thousands of frames, so systematic sub-ulp biases become trajectory drift. `rope_2d_torch` reproduces two kernel-specific behaviours exactly: (i) the frequency-scaling factor F_0 is folded into the inverse frequencies, so that $F_0 = -1$ performs the inverse rotation used by the paged-KV path to un-rotate cached keys, and (ii) every intermediate is rounded to the token dtype at the same points the kernel rounds. The result is `torch.equal`-exact against the repository’s parity reference for `fp32`, `fp16` and `bf16`, verified by CPU-side unit tests; the round trip $\text{RoPE}(F_0=-1) \circ \text{RoPE}$ recovers the input to 2×10^{-7} . Figure 3 shows where the two implementations must agree.

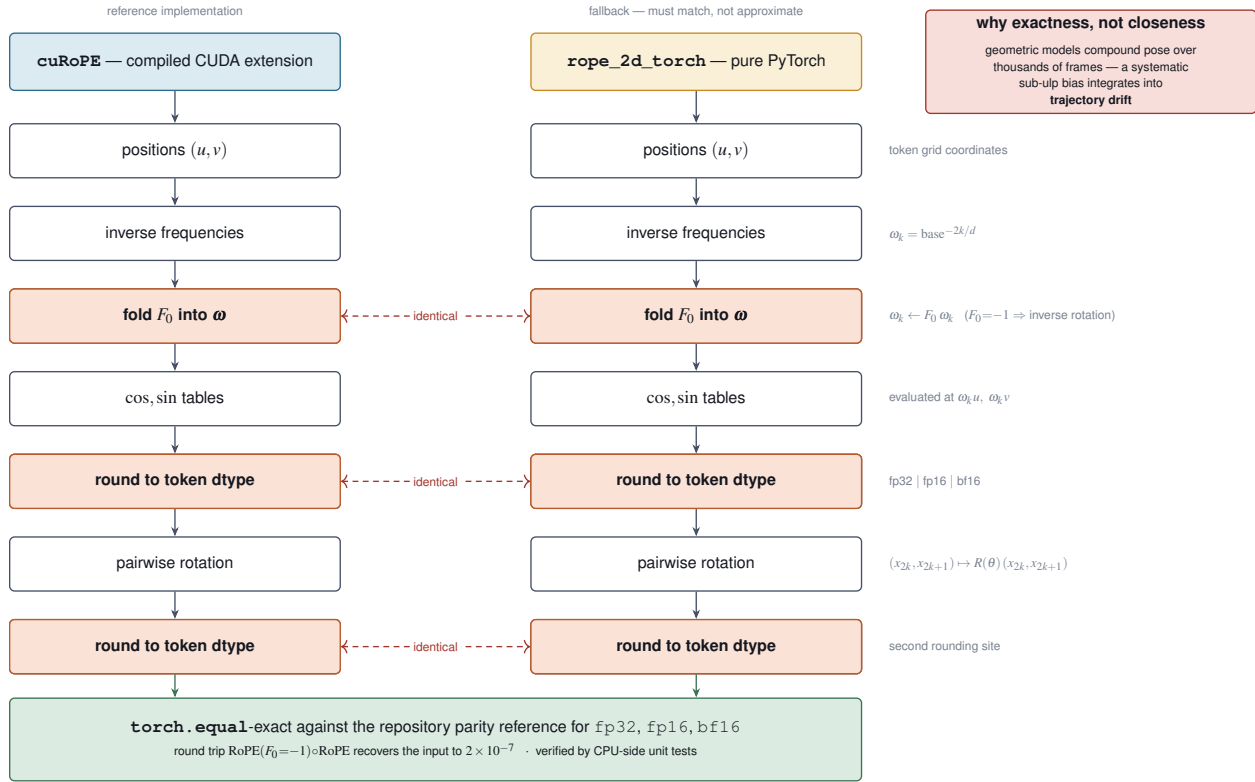


Figure 3: The RoPE fallback matches the kernel’s rounding schedule, not merely its formula. A generic PyTorch rotary encoding would be numerically *close* to the compiled kernel. That is not sufficient here: the two shaded rounding sites and the frequency-scaling fold are the places where a plausible reimplemention silently diverges, and the divergence integrates into pose drift over a long stream. `rope_2d_torch` reproduces both behaviours exactly, which is what licenses the model to run on machines that have no compiled extension at all.

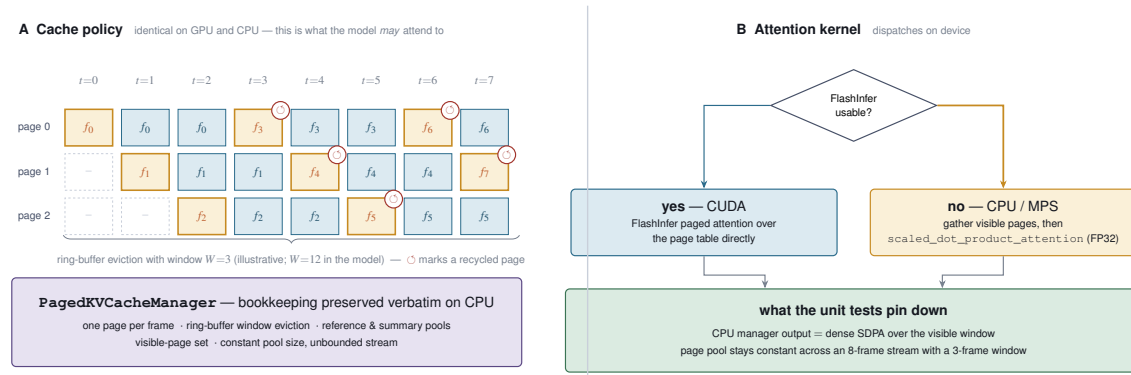
DESIGN RULE. A fallback for a geometry model must match the kernel’s rounding schedule, not just its formula.

3.2 Attention: device-aware backend resolution and paged-KV degradation

Blocker. Backend selection consulted only FlashInfer availability, and the paged KV-cache manager refused to construct without FlashInfer — but FlashInfer’s kernels are CUDA-only regardless of whether the wheel installs.

Fix. Resolution takes the device: on non-CUDA devices, `auto` resolves to PyTorch SDPA, and an *explicit* paged request fails loudly, because silent fallback would misreport benchmarked configurations. Independently, `PagedKVCacheManager` now degrades internally: when FlashInfer is unusable it keeps all of its paging semantics — one page per frame, ring-buffer window eviction, reference and summary pools — and swaps only the attention arithmetic for a gather of visible pages plus `scaled_dot_product_attention` in FP32. Unit tests verify that the CPU manager’s output equals dense SDPA over the visible window, and that page recycling keeps the pool constant across an 8-frame stream with a 3-frame window (Fig. 4).

This separation matters operationally. The *bookkeeping* — what the model may attend to, the fixed civic-memory window — is policy; the *kernel* — how fast attention runs — is capability. Fo Guang wants the policy identical on every tier of hardware and the kernel free to vary.



Separate cache policy from attention kernel. The bookkeeping — what the model may attend to — is *civic policy* and must be identical on every tier; only the arithmetic is capability, and may vary.

Figure 4: Paged KV attention degrades without changing what the model sees. Panel A shows the cache policy the manager enforces regardless of device: one page per frame, ring-buffer eviction of the oldest page once the window is full, and a pool whose size never grows with stream length. Panel B shows the only part that dispatches — when FlashInfer is unusable the manager keeps every paging semantic and swaps the arithmetic for a gather of the visible pages plus `scaled_dot_product_attention` in FP32. Operationally this separation is what lets a donated laptop and a data-centre GPU be governed by the same window policy.

DESIGN RULE. *Separate cache policy from attention kernel; make the policy device-invariant and let only the kernel dispatch on device.*

3.3 Device-agnostic mixed precision

Blocker. Five code sites hardcoded `torch.amp.autocast(device_type="cuda", ...)` — in the streaming frame-forward paths, the Pi3 forward and the camera head — and `torch.cuda.empty_cache()` ran unconditionally after checkpoint load. These are the classic “works on my A100” assumptions.

Fix. Autocast device types derive from the tensors at hand (`hidden.device.type, feat.device.type`); CUDA cache maintenance is guarded by `torch.cuda.is_available()`. At the wrapper level, autocast is enabled only when the device is CUDA *and* the compute dtype is not FP32, so CPU inference runs in clean FP32 without a no-op autocast context, and input frames are materialised in FP32 on CPU rather than bf16.

DESIGN RULE. *Derive device types from data, never from configuration defaults; precision policy is a function of (device, dtype), not a constant.*

3.4 Result: end-to-end CPU inference

With these three families of fixes, `ABotRecon.from_pretrained(checkpoint, device="cpu")` loads the released 4 GB checkpoint and streams video end-to-end with no CUDA runtime, no FlashInfer and no compiled extension. Table 1 reports measurements on 12 real frames of TUM freiburg1_desk [9] at 504×280 on 8 cores of an Intel Xeon Platinum 8559C.

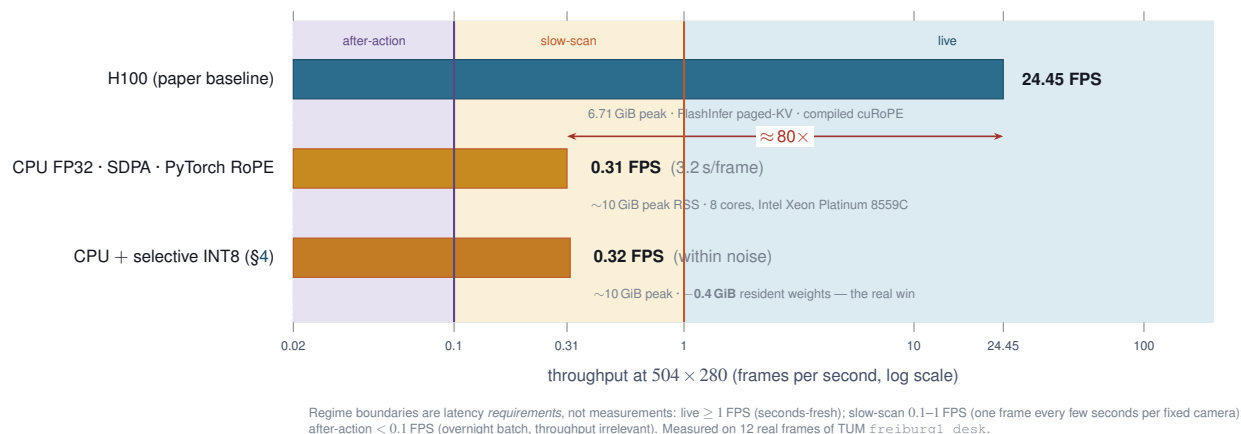


Figure 5: The honest number, and the number that matters. The CPU path is roughly $80\times$ slower than the H100 baseline — and lands squarely inside the latency budget of the two regimes that dominate civil protection. A capability that requires shipping GPUs into a disaster area is a procurement programme; a capability that runs at 3.2 s per frame on the box already bolted next to the camera is a software update. Selective INT8 buys memory, not speed (Section 4).

Table 1: End-to-end throughput and peak resident memory. The CPU rows are measured; the H100 row is the published baseline [1]. Figure 5 places these numbers against the latency budget of each response regime.

Configuration	Throughput	Peak RSS	Notes
H100 (paper baseline)	24.45 FPS	6.71 GiB	FlashInfer paged-KV, compiled cuRoPE
CPU FP32, SDPA, PyTorch RoPE	~0.31 FPS	~10 GiB	3.2 s/frame, 8 cores
CPU + selective INT8 (Section 4)	~0.32 FPS	~10 GiB	~0.4 GiB less resident weights

We state the honest number prominently: **the CPU is $\sim 80\times$ slower than the H100**. [Section 5](#) shows why this is nevertheless the number that matters.

4 Selective Dynamic INT8: Quantisation with a Geometric Conscience

Dynamic INT8 quantisation (`torch.ao.quantization.quantize_dynamic` on `nn.Linear` [8]) is the standard low-effort CPU optimisation, and for classification or language models it is usually benign. For a geometry-regression model it is not, and the failure is invisible unless you measure the *geometry* rather than the loss.

We ablated quantisation scope submodule by submodule against the FP32 CPU baseline on the same 12-frame sequence, reporting trajectory error (ATE against the FP32 trajectory, whose total length is 63 mm), rotation deviation, and relative L2 on dense outputs. [Table 2](#) gives the measurements; [Fig. 6](#) places each submodule on the model’s parameter axis, and [Fig. 7](#) plots the cliff.

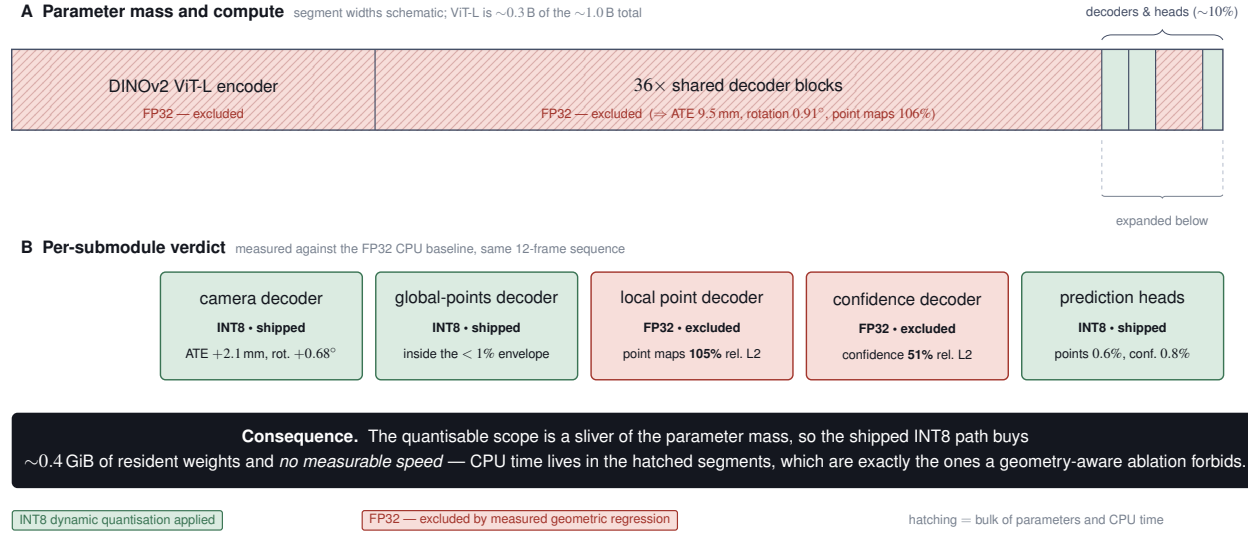


Figure 6: Quantisation sensitivity follows output dynamic range, not parameter count. Panel A places every submodule on a schematic parameter axis; panel B records what happens to the geometry when each is admitted to the INT8 scope. The two facts sit in tension: the modules that dominate parameters and CPU time (hatched) are the ones a geometry-aware ablation forbids, while the modules that may be quantised safely carry almost no mass. That tension is the whole reason the honest benefit reported in Table 2 is memory rather than throughput.

Table 2: Per-submodule dynamic INT8 ablation against the FP32 CPU baseline. Scopes are cumulative. Green marks the sub-1% dense-output envelope of the shipped scope; red marks the two catastrophic transitions; bold in the last row marks the only trajectory regression. The first row is the shipped configuration, whose exact membership is enumerated at the end of this section.

Quantised scope	ATE mean	Rotation mean	Point maps rel. L2	Confidence rel. L2
camera decoder + all heads (<i>shipped</i>)	2.1 mm	0.68°	0.6%	0.8%
+ confidence decoder	2.1 mm	0.68°	0.6%	51%
+ point decoder	2.1 mm	0.68°	105%	51%
+ shared decoder blocks (36)	9.5 mm	0.91°	106%	51%

Three observations follow.

1. **The cliff is per-submodule, not gradual.** Quantising the point decoder alone moves the dense point maps by $\sim 100\%$ relative L2 — the reconstruction is destroyed — while leaving the *trajectory* untouched, because the camera path flows through a different decoder. A pipeline that validated only ATE would ship a model that navigates correctly through geometry that no longer exists. This is not a hypothetical: ATE is the headline metric of the field, and it is precisely the metric that stays clean here.
2. **Sensitivity follows output dynamic range, not parameter count.** The 36 shared decoder blocks — the bulk of parameters and compute — degrade the trajectory modestly; the small point decoder is catastrophic. Per-token activation ranges in the point pathway span the scene's full metric depth, and dynamic per-tensor INT8 cannot represent that range at the precision dense regression requires.
3. **The honest benefit is memory, not speed.** With the shipped scope, throughput is unchanged within noise ($0.31 \rightarrow 0.32$ FPS), because CPU time is dominated by the FP32 DINOv2 encoder and shared decoder blocks — exactly the modules that must stay FP32. The win is ~ 0.4 GiB of resident weight memory, which matters on 8 GB edge boxes running alongside an NVR process.

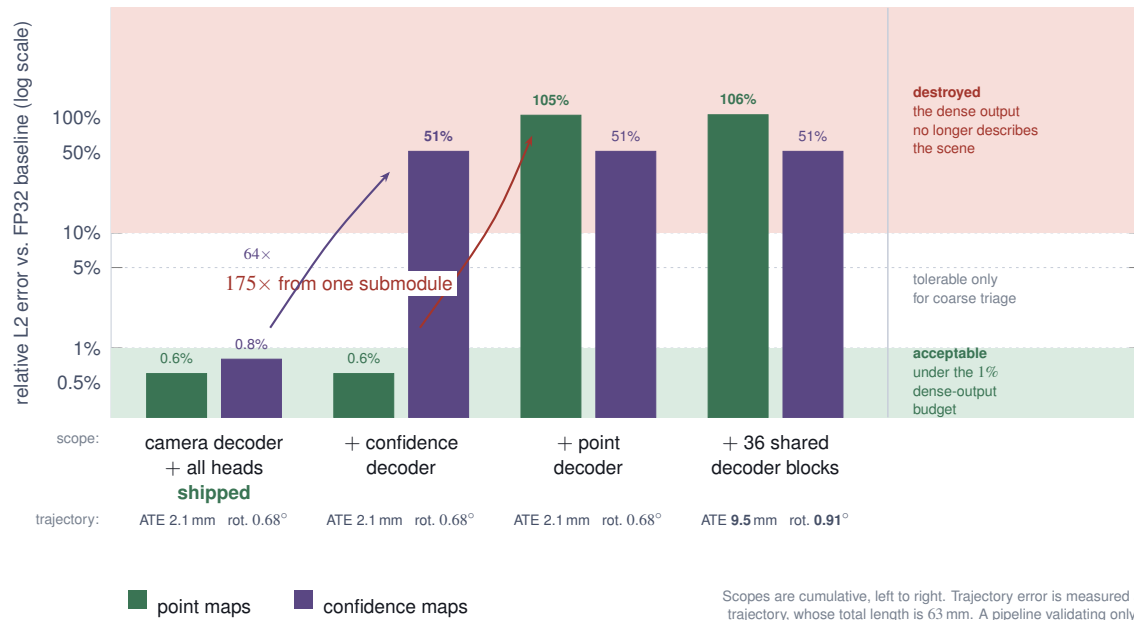


Figure 7: The cliff is per-submodule, not gradual. Admitting the confidence decoder to the INT8 scope multiplies confidence-map error by $64\times$; admitting the local point decoder multiplies point-map error by $175\times$ — while leaving the trajectory numerically untouched, because the camera path flows through a different decoder. The lesson is methodological: quantisation scope for a reconstruction model must be chosen by measured geometric regression per submodule, jointly over trajectory, rotation and dense-output error, and never by a global proxy metric.

The shipped configuration therefore quantises only the camera decoder, the global-points decoder and all prediction heads. It is exposed as `--quantize` in the CLI and `quantization="int8_dynamic"` in the API, is rejected on non-CPU devices, and is recorded in inference metadata — so that downstream consumers, and auditors (Section 6), can see exactly which numeric regime produced a given reconstruction.

METHODOLOGICAL CLAIM. For reconstruction models, quantisation scope must be selected by *measured geometric regression per submodule* — ATE, rotation and dense-output relative error jointly — never by global proxy metrics. We offer Table 2 as a template rather than as a result: the numbers belong to this checkpoint, but the procedure belongs to the model class.

5 Fo Guang: Architecture for Rapid Response

5.1 Three response regimes

Rapid response does not need 24 FPS everywhere. It needs the right latency in the right place (Fig. 8).

- **Live regime** (command posts, drone feeds): seconds-fresh geometry over a handful of priority streams. Served by whatever GPUs the jurisdiction has — a single workstation GPU sustains multiple streams at the model's efficiency point.
- **Slow-scan regime** (fixed-camera canvassing): each of hundreds of fixed cameras contributes a frame every few seconds; scene change, not motion, is the signal. At ~ 3.2 s per frame, *one 8-core CPU box saturates one camera's slow scan* — and every NVR already contains such a box. The CPU path makes the entire installed camera base addressable without new hardware.
- **After-action regime** (damage assessment, insurance, forensics of the disaster itself): overnight batch

	LIVE command posts, drone feeds	SLOW-SCAN fixed-camera canvassing	AFTER-ACTION damage assessment, insurance, forensics
latency budget	seconds-fresh ≥ 1 FPS	one frame every few seconds (0.1–1 FPS)	overnight batch throughput irrelevant
what it answers	is this street passable <i>right now?</i> entry decisions	<i>what changed?</i> scene change, not motion, is the signal	what did the event do to the built environment?
hardware & location	one workstation GPU at the EOC or ground station	8-core x86 NVR / traffic cabinet PC, <i>beside the camera</i>	donated laptops, library PCs, cloud spot CPU — anywhere
configuration	device=cuda FlashInfer paged-KV 24.45 FPS	device=cpu --quantize gather + SDPA; encoder and shared blocks stay FP32 0.31 – 0.32 FPS ≈ 3.2 s/frame	device=cpu correctness and auditability are the whole requirement
addressable base	the handful of streams a jurisdiction can buy GPUs for	the entire installed fixed-camera base one box saturates one camera's slow scan	all recorded footage, at leisure

these two tiers exist only because of the CPU path

Without it, Fo Guang is a proposal to buy GPUs. With it, Fo Guang is a **software update to hardware already deployed** — and, because the reconstruction then runs on the premises, it is also the privacy architecture of [Section 6](#).

Figure 8: Deployment tier map. Rapid response does not need 24 FPS everywhere; it needs the right latency in the right place. The live regime is served by whatever GPUs a jurisdiction already has. The slow-scan and after-action regimes — which between them cover the overwhelming majority of cameras and of the work — are served by commodity processors, and were simply unreachable while the model could not be imported without a compiled CUDA extension.

reconstruction of the day's footage. Throughput is irrelevant; correctness and auditability are everything. Runs on donated laptops, library computers, cloud spot CPU — anything.

The CPU optimisation is what makes tiers two and three exist. Without it, Fo Guang is a proposal to buy GPUs; with it, Fo Guang is a software update to hardware already deployed.

5.2 Reference pipeline

[Figure 9](#) gives the reference deployment. Every stage in it exists in the ABot-Recon repository today — `demo.py`, `scripts/stream_reconstruction.py`, `scripts/export_reconstruction_ply.py` and the GEV consumer. Fo Guang's contribution at this layer is the *deployment mapping* — which regime runs where — plus the governance boundary of [Section 6](#). The fixed 12-frame context window gives every tier identical, bounded memory behaviour: a camera that streams for a month costs the same RAM as one that streams for a minute.

5.3 Why monocular reconstruction is the right primitive

Rapid response has tried and abandoned heavier primitives. LiDAR maps are precise but stale and sparse in coverage; photogrammetry requires flight operations; classical SLAM stacks require calibration and per-deployment tuning. ABot-Recon's primitive — *uncalibrated monocular video to metric-consistent geometry*

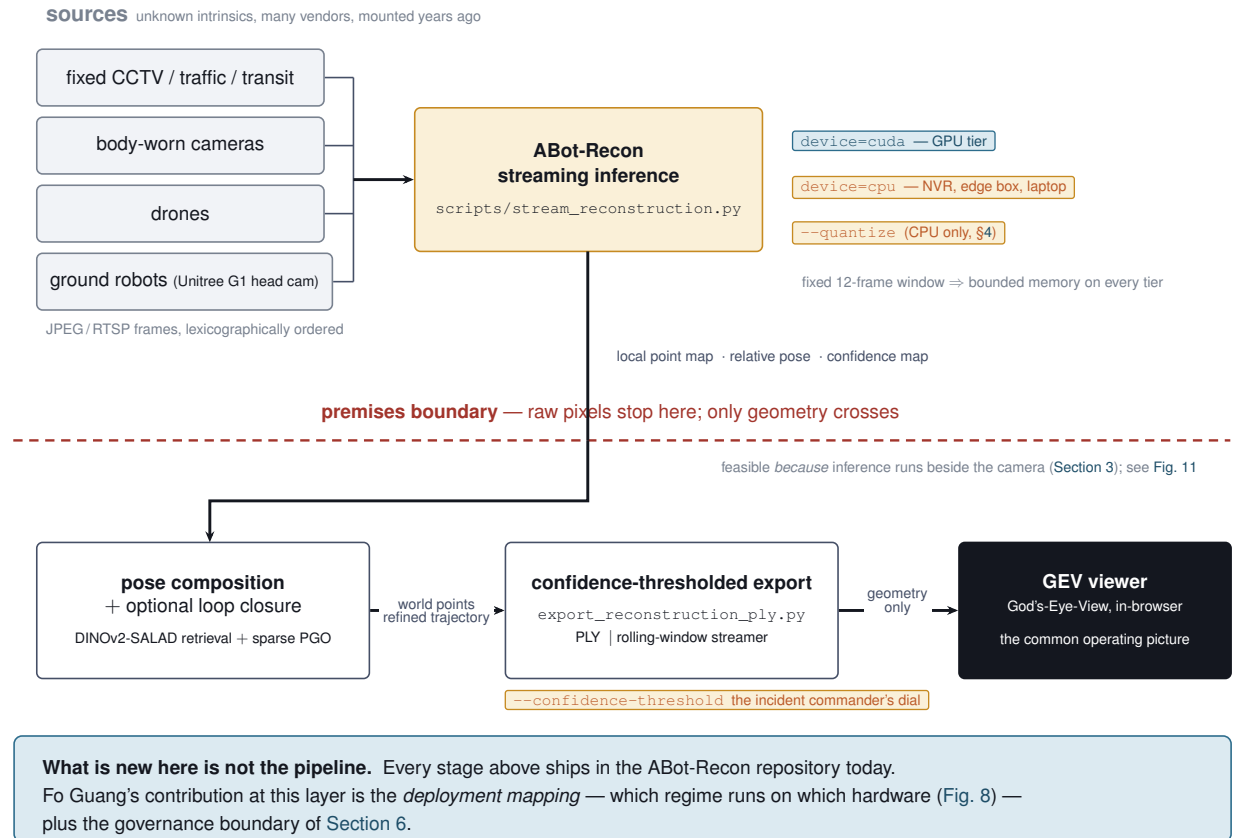


Figure 9: The Fo Guang reference pipeline. Uncalibrated monocular video enters from whatever optics a jurisdiction already owns; metrically consistent geometry with per-pixel confidence leaves. The primitive matters. LiDAR maps are precise but stale and sparse in coverage, photogrammetry needs flight operations, and classical SLAM stacks need calibration and per-deployment tuning — none of which survives contact with a camera inventory of unknown intrinsics mounted by a dozen vendors over fifteen years. The confidence map is operationally essential rather than decorative: responders must know which parts of the model to trust before committing entry teams.

with per-pixel confidence — matches the actual inventory: cameras of unknown intrinsics, mounted years ago, by many vendors.

The confidence map is operationally essential rather than decorative. Responders must know which parts of the model to trust before committing entry teams, and `--confidence-threshold` gives incident commanders a single dial from “show everything” to “show only what the model is sure of.” A reconstruction system for emergency use that reported geometry without calibrated uncertainty would be worse than no system, because it would be trusted at the wrong moments.

6 Governance: Light, Not Gaze

A system that ingests every public camera is a hazard as much as an asset. Fo Guang's governance design rests on one technical fact and three policy mechanisms.

The technical fact: geometry is not identity. The Fo Guang egress is point clouds, trajectories and confidence maps at 280×504 . This output class does not carry faces, gait signatures, licence plates or re-identifiable appearance; the reconstruction of a street tells you that the street collapsed, not who was standing on it. By fixing the *output type* at the architecture level — the model has no identity heads to enable

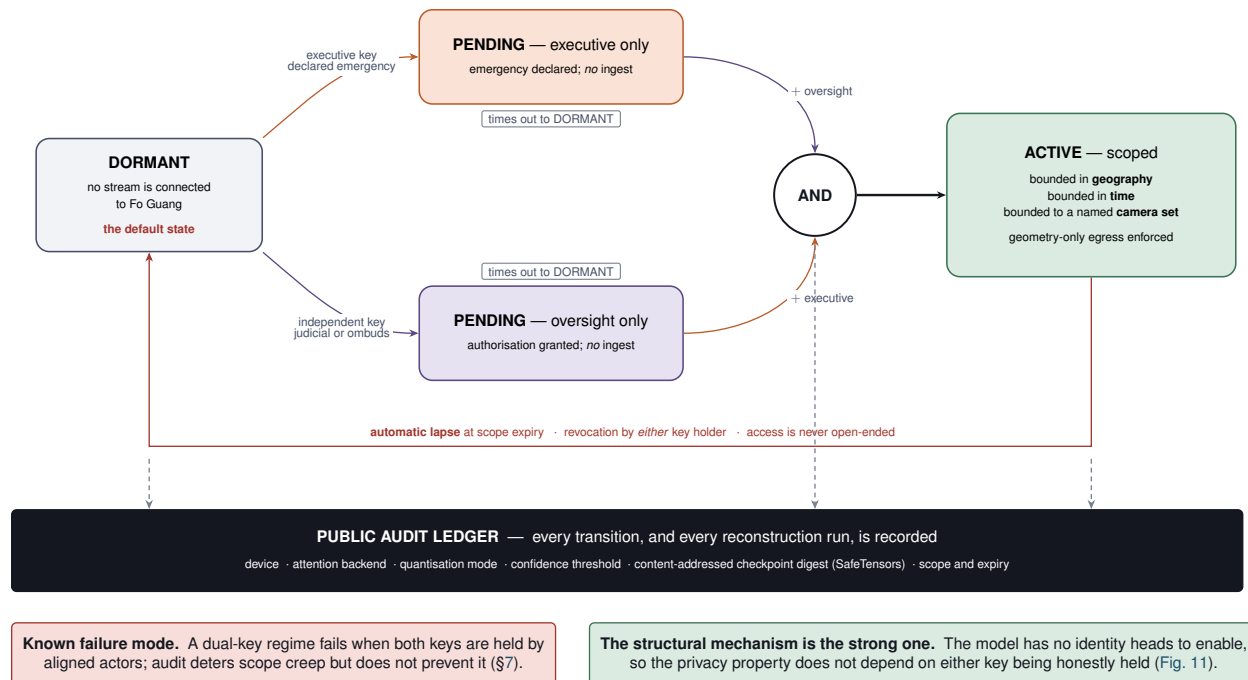


Figure 10: Dual-key activation: the default state is off. Camera streams are not connected to Fo Guang by default. Ingest requires a declared emergency *and* an independent authorisation, scoped in geography, time and camera set, after which access lapses automatically rather than persisting until someone remembers to revoke it. Either key holder alone reaches a pending state that grants nothing. Every transition is written to a public ledger together with the numeric regime that produced each reconstruction, so an auditor can tell not only *that* the system ran but under exactly which configuration.

— the privacy property is structural rather than promised (Fig. 11).

Mechanism 1: dual-key activation. Camera streams are not connected to Fo Guang by default. Ingest requires a declared emergency (executive key) *and* an independent authorisation (judicial or ombuds key), scoped in geography and time, after which access lapses automatically. The default state is off. Figure 10 makes the state machine explicit, including the two pending states that grant nothing and the automatic lapse that removes the need for anyone to remember to revoke.

Mechanism 2: geometry-only egress at the edge. Reconstruction runs *on the NVR or edge box beside the camera* — feasible precisely because of the CPU path — and only geometry leaves the premises. Raw pixels never traverse the network to a central authority. The CPU optimisation is thus not merely a cost story; it is the *privacy architecture*: computation moves to the data so that imagery does not move to the state.

Mechanism 3: public audit of the pipeline itself. Inference metadata records device, attention backend, quantisation mode and confidence threshold per run; checkpoints are content-addressed via SafeTensors digests; the model, the tests and this paper’s ablation are public. Anyone can verify that the deployed artefact computes geometry and nothing else, and reproduce its accuracy envelope on a laptop — because it runs on a laptop. That last clause is doing real work: an audit right that requires an H100 to exercise is an audit right held by institutions that already own H100s.

We name the project for this inversion. Fo Guang (Fó Guāng), “Buddha’s light,” is illumination that falls on everyone and singles out no one. The same optics that constitute a surveillance state when pointed at *persons* constitute a public good when constrained — technically and institutionally — to *places*.

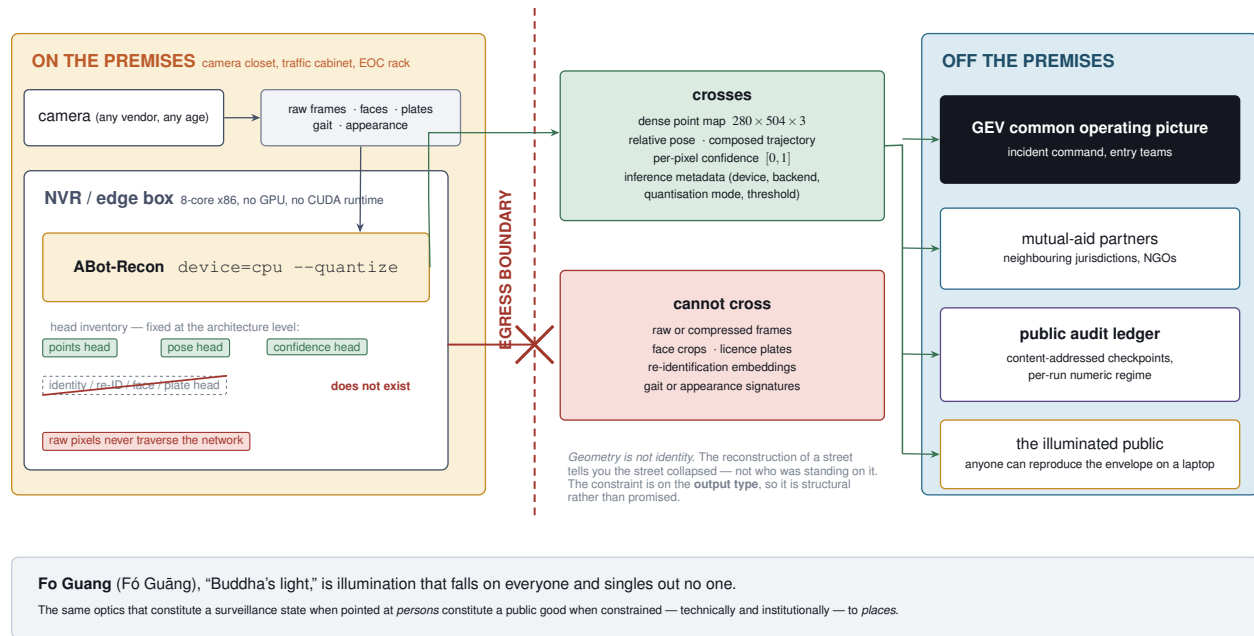


Figure 11: Geometry-only egress at the edge. Reconstruction runs on the box beside the camera — feasible precisely because of the CPU path of Section 3 — and only geometry leaves the premises. The CPU optimisation is therefore not merely a cost story: it is the privacy architecture, because computation moves to the data so that imagery need not move to the state. Of the three governance mechanisms, this is the one that does not depend on anybody’s good faith, and the one we think deserves formalisation (for example, attested edge binaries).

7 Limitations and Future Work

- **Throughput honesty.** 0.31 FPS on 8 cores serves the slow-scan and after-action regimes, not live video. Closing the gap on CPU likely requires INT8 or bf16 kernels for the encoder and shared decoder blocks with quantisation-aware finetuning — our ablation shows that post-training dynamic INT8 cannot reach them safely.
- **Single-sequence evaluation.** The accuracy deltas of Section 4 are measured on one 12-frame indoor sequence. A civic deployment demands that the ablation be re-run across the model’s evaluation suites (KITTI, TUM, ETH3D-class data) and across seasons and weather for fixed cameras. We report the procedure with more confidence than the constants.
- **Multi-camera fusion.** Fo Guang currently reconstructs per stream and fuses in the viewer; principled cross-camera registration — a shared loop-closure graph across the network — is the natural next system, and the one that would turn many local reconstructions into a single civic map.
- **Adversarial governance.** Dual-key regimes fail when both keys are held by aligned actors, and the audit mechanism deters but does not prevent scope creep. We consider the structural output-type constraint the strongest of the three mechanisms and the one deserving formalisation — for example, attested edge binaries whose measured hash is bound to a published model with no identity heads.

8 Conclusion

The surveillance state’s cameras are a sunk cost; its harms come from what is computed on them. ABot-Recon’s CPU path — a bit-identical RoPE fallback, device-aware attention with a gracefully degrading paged-KV cache, device-agnostic precision handling, and a geometrically audited selective INT8 mode —

demonstrates that billion-parameter reconstruction now runs on the commodity processors already wired to those cameras. That single systems fact reorders the possibility space: geometry can be computed where the pixels live, identity need never leave the building, and the output — a live map of a stricken city — belongs to everyone it illuminates.

Turning the surveillance state into a public good is not a metaphor awaiting technology. As of the CPU path, it is a deployment decision.

Artefact Availability

The CPU inference path, the selective INT8 mode and the per-submodule ablation harness reproduced in [Table 2](#) are public in the repository cited as [10]. The accuracy envelope of [Tables 1](#) and [2](#) is reproducible on any 8-core x86 machine with 16 GB of RAM — deliberately, since an audit right that requires a data-centre GPU is not a public audit right.

References

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- [9] J. Sturm, N. Engelhard, F. Endres, W. Burgard, D. Cremers. A benchmark for the evaluation of RGB-D SLAM systems (TUM RGB-D). *IROS*, 2012.
- [10] This repository: CPU inference and selective INT8 path, PR #4 (`gratitude5dee/ABot-Recon`), including the per-submodule quantisation ablation reproduced in [Section 4](#).